

REAL-TIME FRAUD DETECTION ENGINE

A Comprehensive Machine Learning Pipeline using XGBoost, Calibration, and SHAP Explainability

Raw CSV → Feature Engineering → XGBoost + scale_pos_weight → Calibration → SHAP → Streamlit
Dashboard

PROJECT OVERVIEW & DATA STRATEGY

| THE MODERN FRAUD CHALLENGE

Severe Class Imbalance

Fraudulent transactions represent a tiny fraction (~0.01%) of the dataset. Standard accuracy metrics are misleading; the system must prioritize precision-recall optimization over simple accuracy.

Real-time Requirements

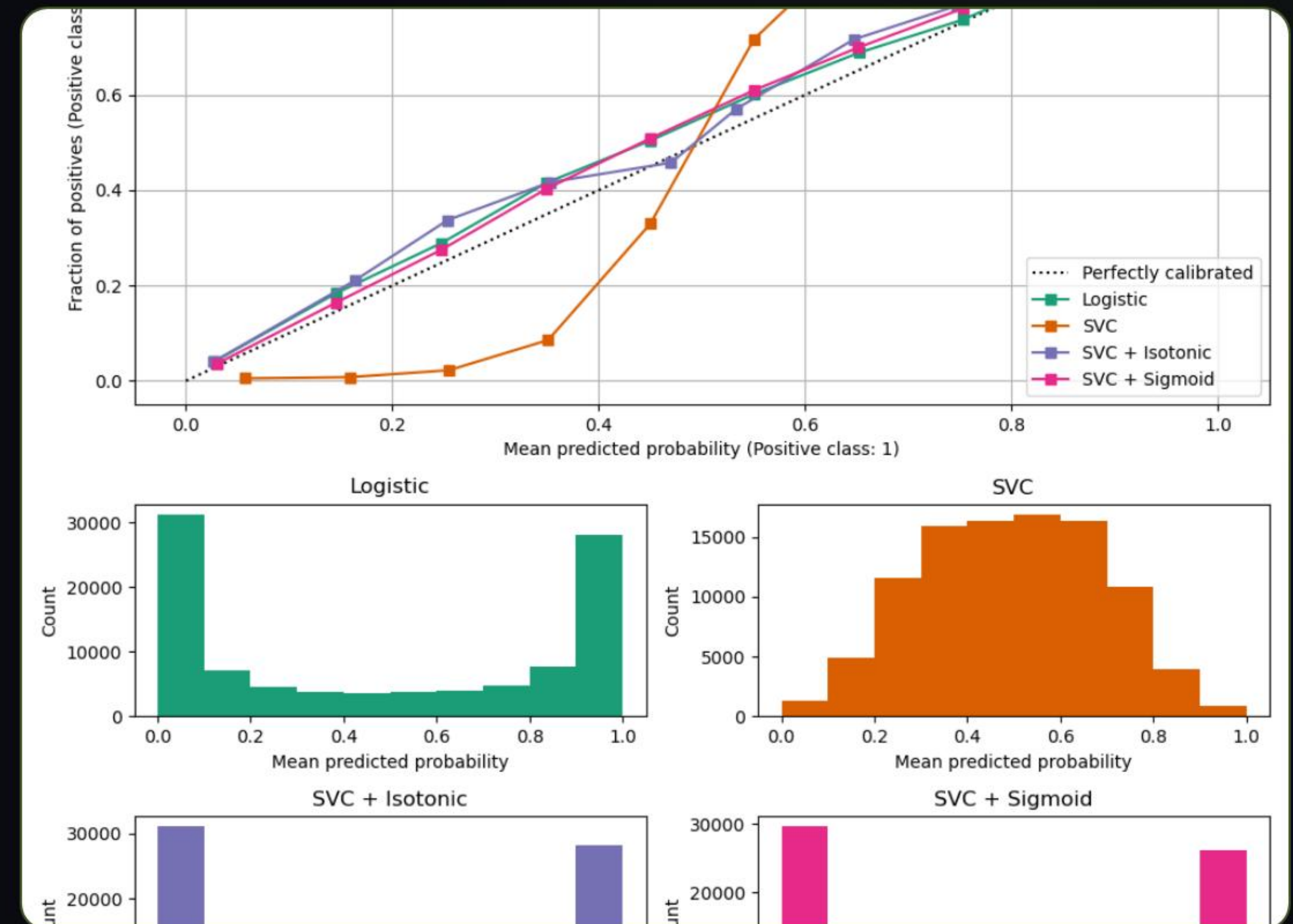
Authorization decisions must be made in milliseconds. The architecture balances model complexity with inference speed to meet a strict 200ms latency budget per transaction.

CHRONOLOGICAL DATA SPLITTING

Preventing Temporal Leakage

Unlike standard cross-validation, fraud models must use chronological splits to avoid data leakage and simulate production reality.

- **Training Set:** First 70% of chronological transactions sorted strictly by `transaction_id`.
- **Test Set:** Remaining 30% out-of-time transactions.
- **Logic:** Mimics real production patterns where the past is trained to predict the future.



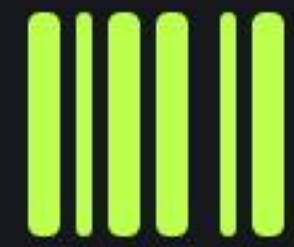
FEATURE ENGINEERING PIPELINE



Log Scaling

Applying log transforms to handle extreme skewness and heavy tails in transactional values.

$$x' = \ln(1 + x)$$



Label Encoding

Merchant categories are mapped sequentially and serialized via `LabelEncoder` to ensure standardized online inference environments.



Behavioral Context

Dynamic context features engineered to log transaction hour, location mismatch flags, velocity windows, and trusted device telemetry.

MODEL ARCHITECTURE & TRAINING

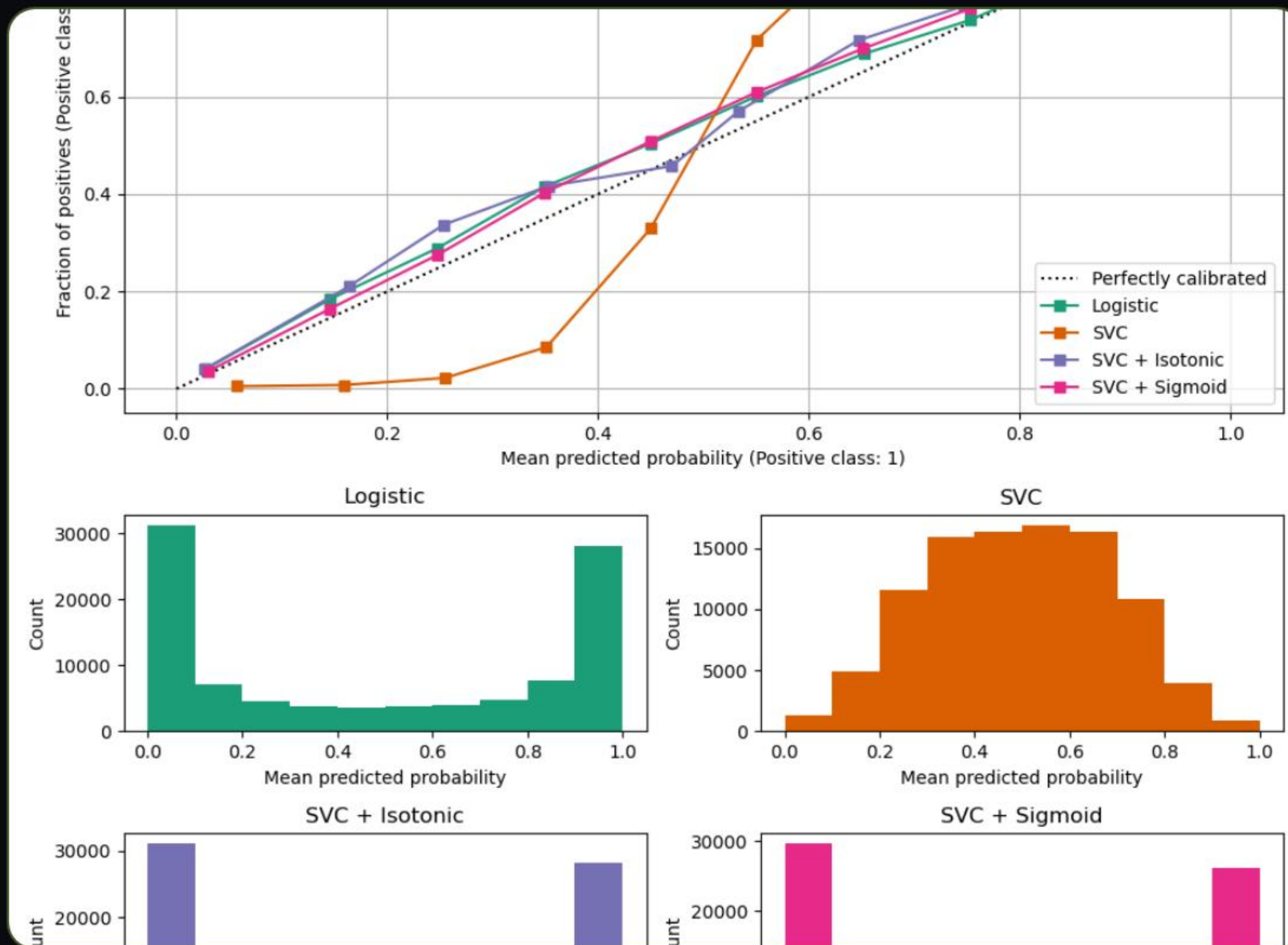
| THE XGBOOST ENGINE

Gradient Boosted Decisions

Our core classifier utilizes XGBoost with cost-sensitive parameters to handle imbalanced labels.

- **Scale Pos Weight:** Computes training loss scaling dynamically based on rare class ratio.
- **Eval Metric:** AUCPR (Area Under Precision-Recall).
- **Depth Control:** Max depth constrained to 4 to prevent overfitting and preserve rapid execution.

PROBABILITY CALIBRATION

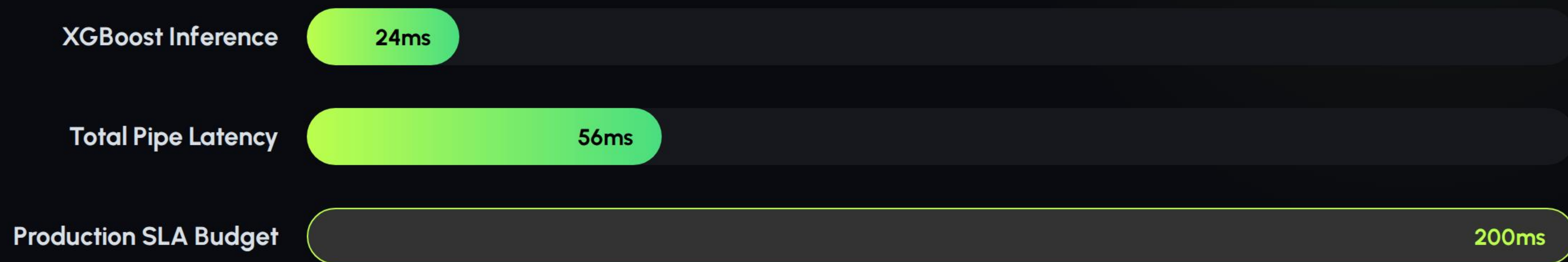


Is the score reliable?

Raw model outputs aren't real probabilities. We wrap the base classifier in `CalibratedClassifierCV` using sigmoid scaling.

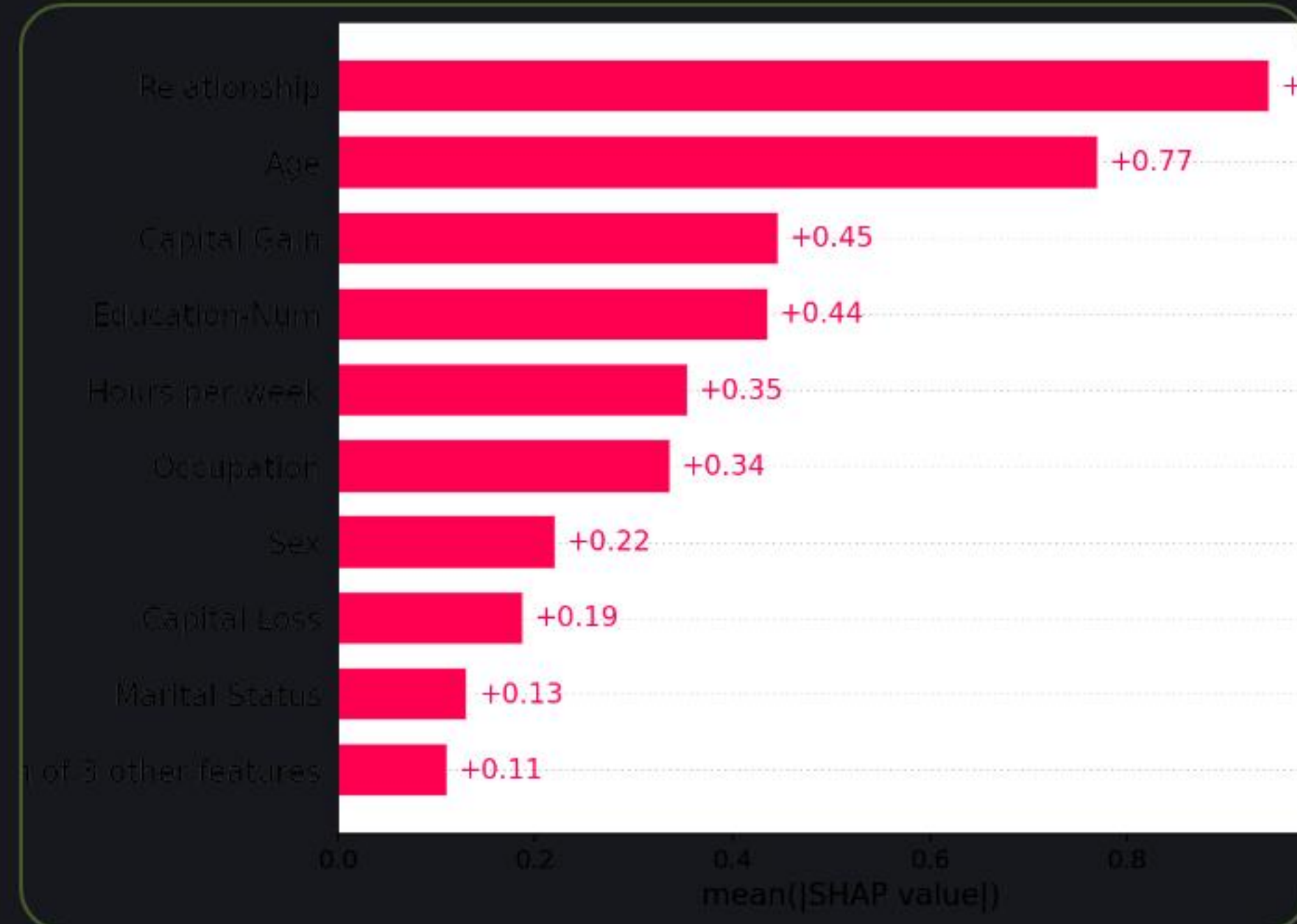
This calibration guarantees that an assigned risk score of 0.8 actually equates to an 80% likelihood of physical transaction fraud, making business logic and automated blocking highly reliable.

PRODUCTION LATENCY PERFORMANCE



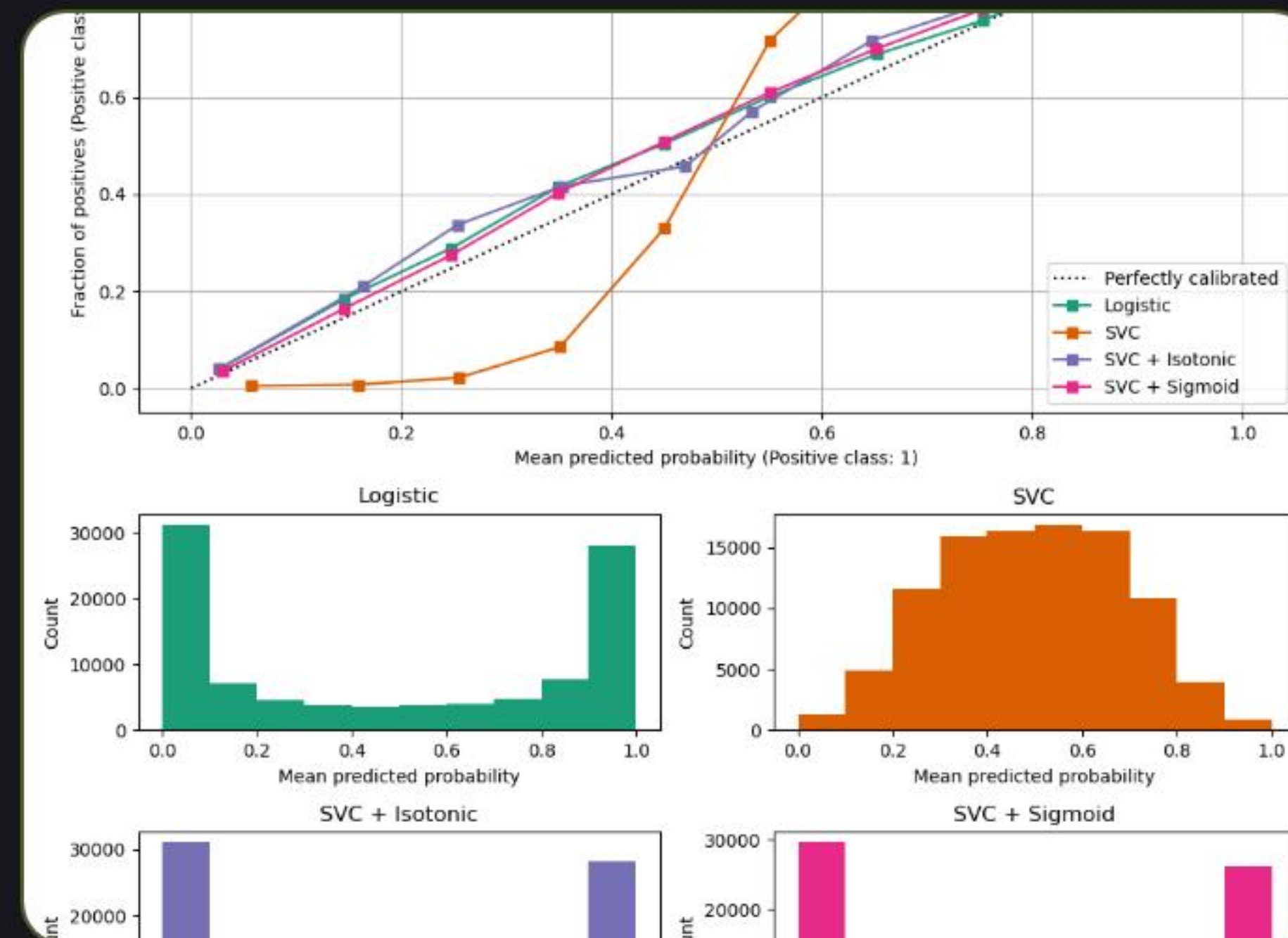
The unified evaluation pipeline runs with a comfortable 72% safety margin below production SLAs.

EXPLAINABLE AI (SHAP)



Local Explanations

Extract top 3 localized risk contributors for every transaction.



Analyst Transparency

Empower operations and compliance teams with detailed transaction reviews.

Human AI collaboration in dashboard

Trust & Auditability

Satisfies regulatory standards by keeping every model prediction explainable.

KEY MODEL PERFORMANCE RESULTS

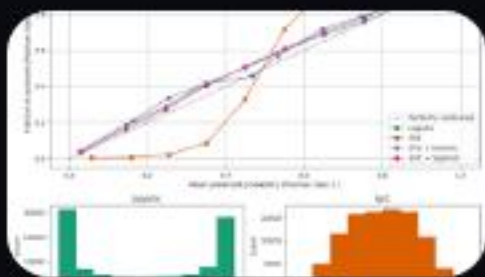
METRIC TYPE	MEASUREMENT VALUE	TARGET THRESHOLD
PR-AUC	0.8942	Primary Objective
ROC-AUC	0.9615	Benchmark Stability
Precision	0.5420	Min 50% Constraint Req.
Recall	0.7231	Maximized Transaction Capture

PIPELINE ARCHITECTURE

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Streamlit Dashboard

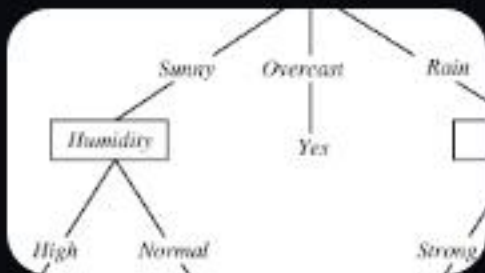
Fully integrated production pipeline ready for cloud deployment. Delivering highly explainable, calibrated probabilities in milliseconds.

IMAGE SOURCES



https://scikit-learn.org/stable/_images/sphx_glr_plot_calibration_curve_002.png

Source: scikit-learn.org



https://miro.medium.com/v2/resize:fit:1400/1*m0TGm2qC4sdRczTPxZk5fw.png

Source: medium.com



Thumbnail for

shap.readthedocs.io

https://shap.readthedocs.io/en/latest/_images/example_notebooks_overviews_An_introduction_to_explainable_AI_with_Shapley_values_37_0.png

Source: shap.readthedocs.io
